

GVVN 2021 Flight Activity

Applied statistics

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1 Introduction

(To be added in next report)

1.1 Research questions

1.2 Approaches

2 Dataset

The dataset for this project comes from the GVVN and lists all glider flights performed in 2021 within the club. For each of these, 19 attributes (variables) are given and explained below:

- **Date** : date of the flight (always in 2021)
- **Jour** : day of the flight
- **No** : will not be used (information for billing purposes)
- **Nom pilote** : name of the pilot (anonymized)
- **Moniteur** : name of the flight instructor, second pilot (anonymized) or passenger (denoted as "PAX")
- **Pilote remo.** : name of the towing pilot (anonymized)
- **GV** : type of flight;
 - VP: private flight
 - VEC: school flight
 - VFO: training, refresher or check flight
 - VCR: check flight for tow pilot
 - VTT: transition flight (switching to a new glider)
 - VFP: private training flight
 - VPP: private glider flight (GVVN member)
 - BON: flight voucher
 - EXT: external flight (GVVN non-member)
- **Remo** : tow plane registration;
 - WYZ: ultralight sport aircraft (Flight Design CTLS)
 - 2451: touring motor glider (Samburo AVO68-R 115)
- **Planeur** : glider registration;
 - two-seaters: 3354, 3185, 2492 (turbo)
 - single-seaters: 1840, 1636, 2509 (turbo), 3147 (one flight, sold afterwards)
 - vintage gliders: 653, 552
- **H décoll.** : takeoff time
- **H att. remo.** : tow plane landing time (usually earlier than the glider landing time)
- **H att. plan.** : glider landing time
- **Durée remo.** : tow plane flight duration
- **Durée plan.** : glider flight duration
- **Fact** : will not be used (information for billing purposes)

- **Mois** : month of the flight
- **Planeur Club** : GVVN glider registration
- **Planeur privé** : private glider registration
- **1er pilote** : will not be used (information for billing purposes)

In addition, meteorological data for the city of Neuchâtel were added for each flight day (hourly measures with takeoff time as reference). They were extracted from the Meteostat¹ database. Here are the 8 variables used:

- **temp** : temperature (°C)
- **dwpt** : Dewpoint temperature (°C)
- **rhum** : Humidity (%)
- **prcp** : total precipitation (mm)
- **wdir** : wind direction (1-360°)
- **wspd** : wind speed (km/h)
- **wpgt** : peak gust (km/h)
- **pres** : air pressure (hPa)

As indicated, the names of the pilots and instructors have been anonymized (each name has been associated with another random name) for reasons of data protection. Note that the attribution was made regardless of gender. Care must be taken with the type of flight, which can sometimes be incorrect. As some of the codes are quite specific, it can happen that the person in charge of recording the flights makes a small mistake. The club owns two tow planes, the 2451 weighing about 260 kg more than the WYZ. Two gliders are labelled "turbo", which means that they have a retractable turbo sustainer engine² that can be started upwind (i.e. in flight). This provides additional security and guarantees a safe return to the departure airfield, or avoids losing altitude or even permits climbing. Vintage gliders are set apart, as their performance is no longer comparable to modern gliders (for every metre of altitude lost, a modern glider will cover about twice the horizontal distance of an old glider).

To find out more about the GVVN or its fleet, the reader is invited to visit its dedicated website (Groupe de Vol à Voile de Neuchâtel, 2022).

3 General flights analysis

3.1 How does the number of flights or total flown hours per day/month evolve through time?

For this first analysis, let's investigate the most popular time to fly. This will be exclusively done through visualization.

The first thing of interest is the distribution of glider flights across the day. Figure 1 represents the histogram of the hours of gliding. The club's activity starts at 9:00 am and ends at 7:00 pm. We first notice that there is a distinct dip at noon. This is explained by the lunch break.

¹<https://meteostat.net/fr/>

²<https://www.schempp-hirth.com/en/individual/engine-options>

Apart from that, we can notice that the hours distribution is quite similar to the Gaussian distribution concentrated around 2:00 pm.

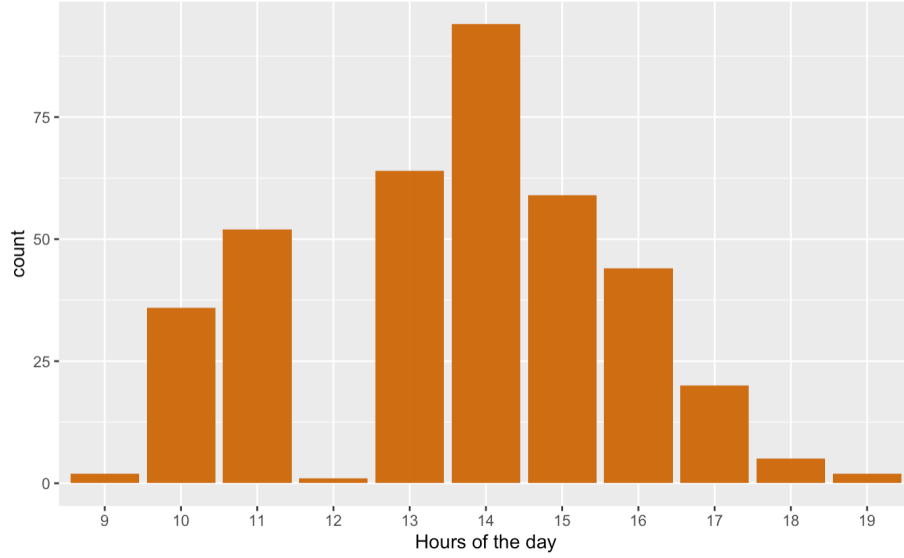


Figure 1: Histogram of distribution of the flight's hours throughout the day. The x-axis represents the hours of the day. The y-axis represent the count. We see aside from the drop of the lunch time, the distribution is unimodal, strongly resembling a normal distribution.

The second aspect analyzed is the distribution of flights across the month. Figure 2 shows that there is no directly apparent trend. This pattern may result from the fact that not all months are similar, indeed some months may contain more vacations than others. This also comes from the fact that weekends do not always occur on the same dates as 7 is not congruent with 29,30 nor 31. However one can see that there is a sharp drop in the middle of the month. Further analysis have to be done to investigate this decrease.

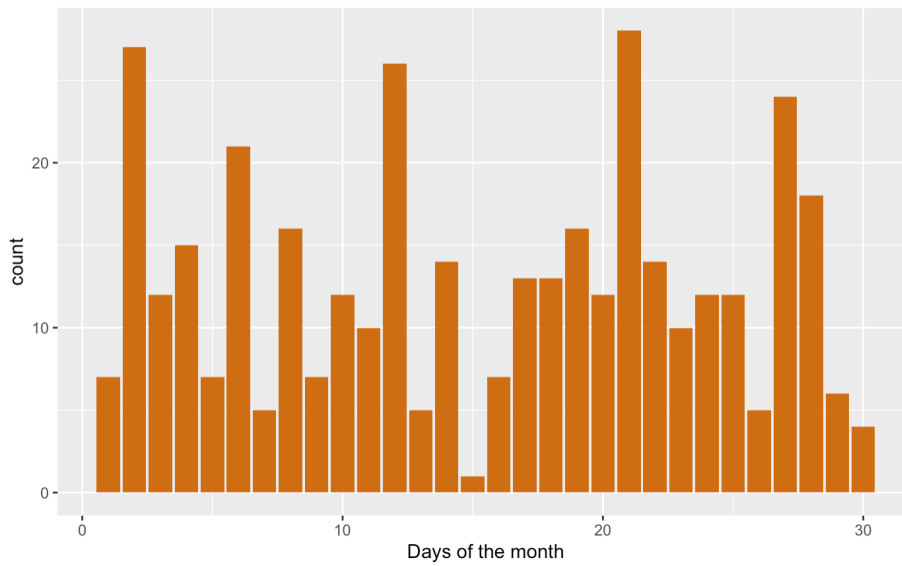


Figure 2: Histogram of distribution of the flight's days throughout the month. The x-axis represents the day of the month. The y-axis represent the count.

To further explore the distribution of flight days, this time let's examine their distribution on the weekdays. Figure 3 shows that Saturday is the most popular day. Notice that there are no flights on Tuesday and Thursday. There is no particular reason for this, except that there were not many vacations on those two days.

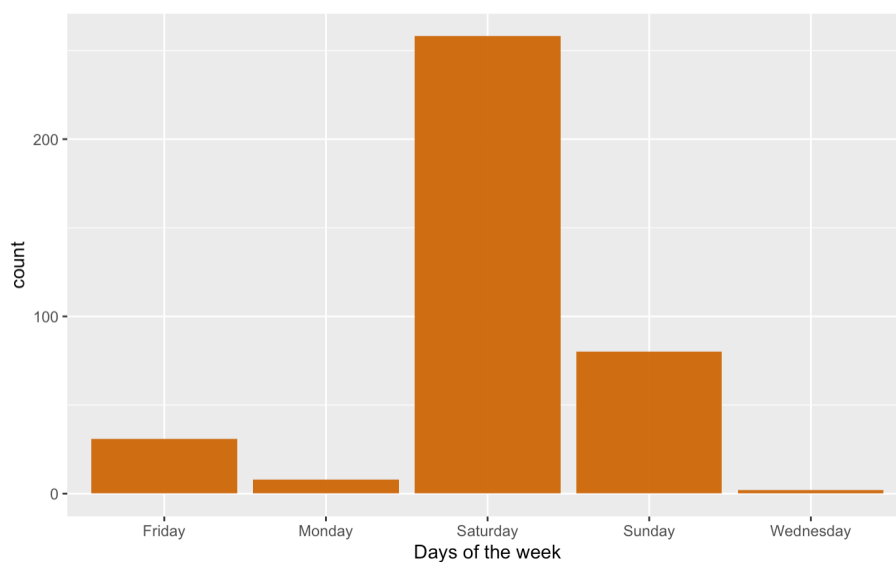


Figure 3: Histogram of distribution of the flight's days of the week. The x-axis represents the days. The y-axis represent the count. Most flights happen on the week-ends, especially on Saturday.

Now let's move on to a more global analysis and look at the distribution of flights over the year. Figure 4 shows that the majority of flights occur in the spring and early fall with March the most popular month. This is to be expected as these are the months with the most favorable weather conditions. In Section 4, we will look at the effect of weather in more detail.

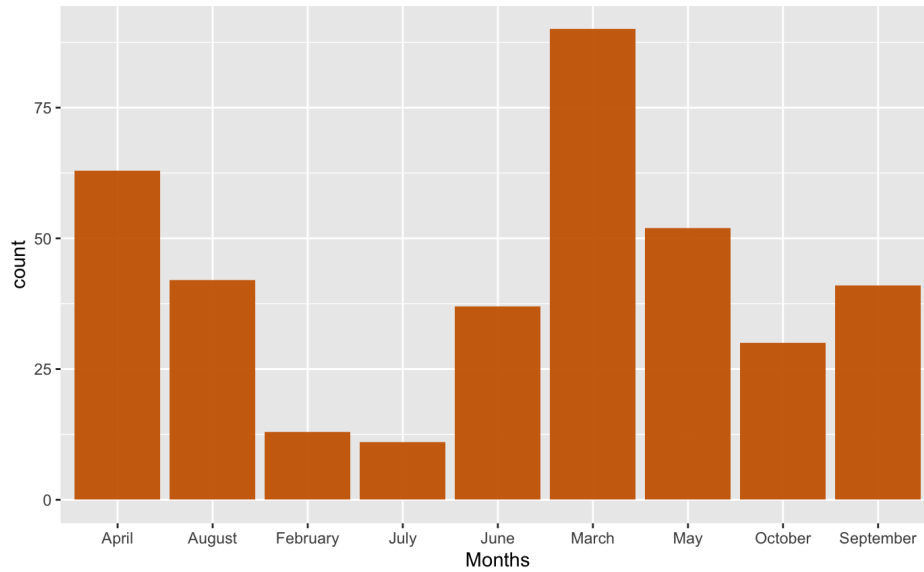


Figure 4: Histogram of distribution of the flight's months. The x-axis represents the months. The y-axis represent the count. Most flights happen in March.

3.2 What aspects in a flight characterizes the flight duration?

Once the overall picture of how flights are distributed over time is known, we intend to investigate the factors that may characterize the flight duration. For this preliminary report, we will investigate two methods, the first one the correlation between the duration and the other parameters. The second will be done through a regression. For this analysis the main focus is on the column `Plan duration.`. This column corresponds to the difference between the column `H décoll.` and `H att. plan.`. Note that this duration also includes the towing time. We are aware that this analysis does not distinguish between long flights and flights that were shorter but went farther from the club and thus had a correspondingly long towing time. In consideration of the difference between takeoff and landing time, we are limited on this issue and continue the analysis despite the nuance.

For this analysis, dummies variables are created for all the factors. We then move from a dataset of dimensions to 379 rows with 19 columns to 124 columns.

Finally for this section, columns `Planeur Club` and `Planeur privé` were merged to avoid missing values. A new columns named `Club` with an indicator of the status of the flight (private or club) is added, where `=1` means a glider from the club and `=0` refers to a private flyer. In Figure 5 is displayed the ratio between the two categories. We can see that only 9.23% of the flight are private whereas the rest is part of the club.

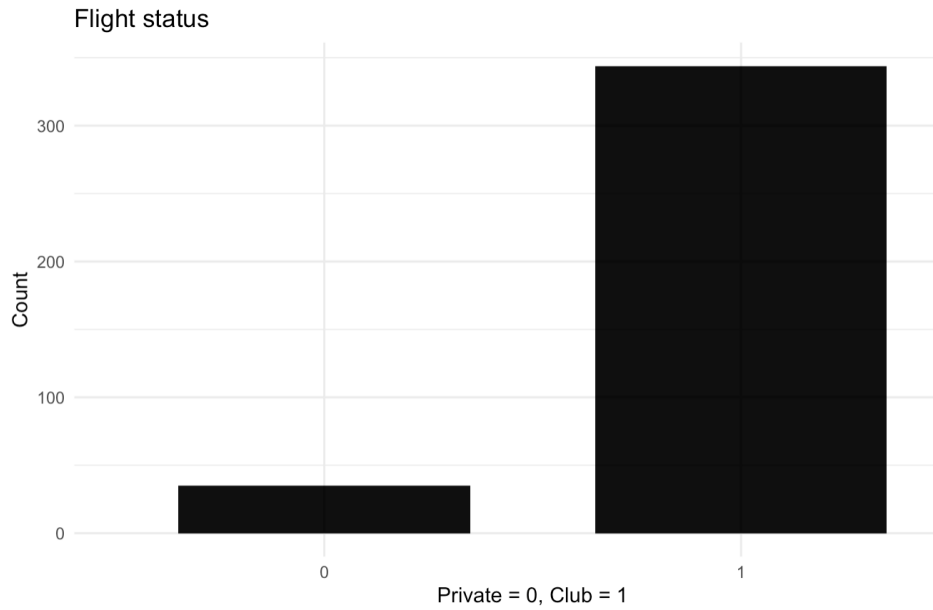


Figure 5: Barplot comparing the number of private to club flights. Club =1 refers to a club flight whereas =0 refers to a private flight. The gliders represented in the dataset are predominantly from the club

3.2.1 General overview of duration

Before starting with the analysis of the eventual factors determining the flight duration, let's look at the distribution of the flight duration. The average flight duration is 62.8 minutes and the median duration is 33 minutes. Figure 6 shows the distribution of durations. The distribution strongly resembles a power law model, making it difficult to identify the mean and median on the plot.

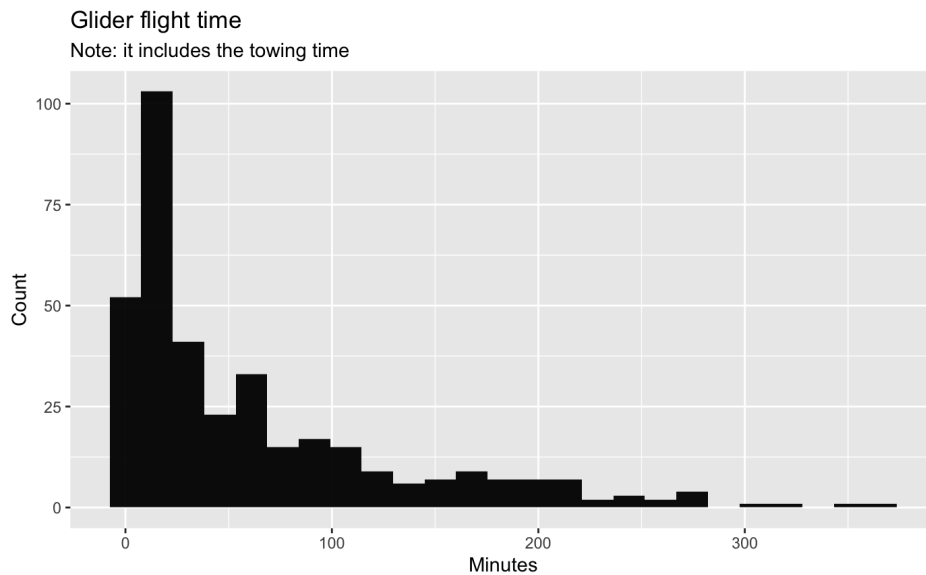


Figure 6: Histogram of distribution of the flight's duration. The x-axis represents the time in minutes. The y-axis represent the count. We can see a power law distribution shape.

3.2.2 Correlation

Let's first look at the correlation between flight time and the rest of the parameters. Figure 7 represents the 20 variables most correlated with the flight duration. In first position we find the status plays a big role, indeed the correlation with the variable **Club** leads us to deduce that private flights tend to last longer. We also see, as expected, that the longer the flight time of the towing vehicle, the longer the flight time of the planner, as this means that the flight location is further away from the club. We also see different factors from the different pilots and planners available in the club. Again, the names are anonymized and do not reflect the identity, gender or origin of the person.

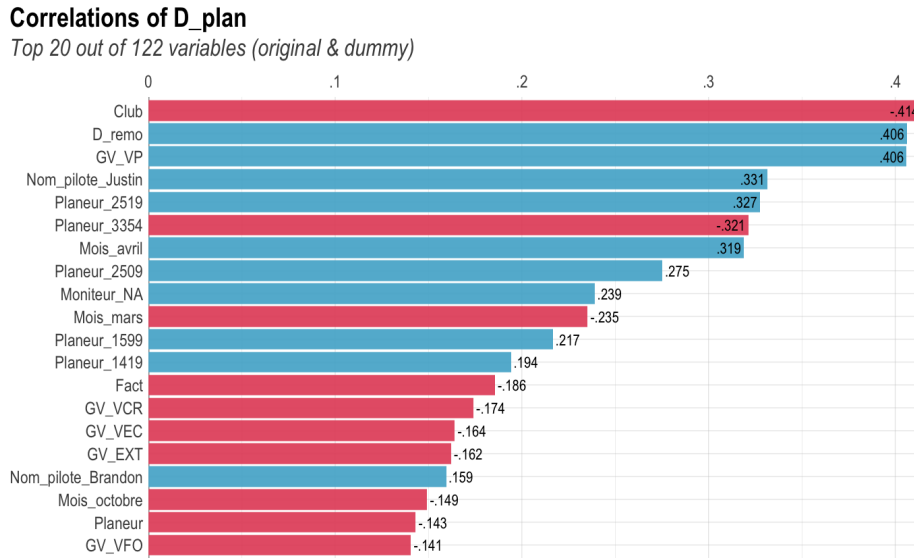


Figure 7: Top twenty variables correlated with the flight duration. The bars in blue represent positive correlation while the one in pink represent negative correlation. The highest correlation has a absolute value of a bit more than 41%.

3.2.3 Linear Regression

To conclude this analysis, for now, we perform a linear regression to identify the different effects of the variables. Because of the large number of variables, we perform a step regression in both directions, forward and backward, based on the AIC¹ criterion. Table 11 found in Appendix A represents the summary results of this regression. We can see that the overall Adjusted R-squared is 0.7 which is relatively good. Looking at the figure, we see that the residuals bounce off the 0 line. This suggests that the assumption that the relationship is linear is reasonable. The residuals form approximately a "horizontal band" around the 0 line. This suggests that the variances of the error terms are equal. Looking at the scale-location, this suggests that the assumption of homoscedasticity is poorly verified. Finally, looking at the normal QQ plot, we can see that the error is very slightly short-tailed.

¹The Akaike information criterion gives a measure assessing the quality of a model based on the goodness of fit and the simplicity (number of independent variables involved).

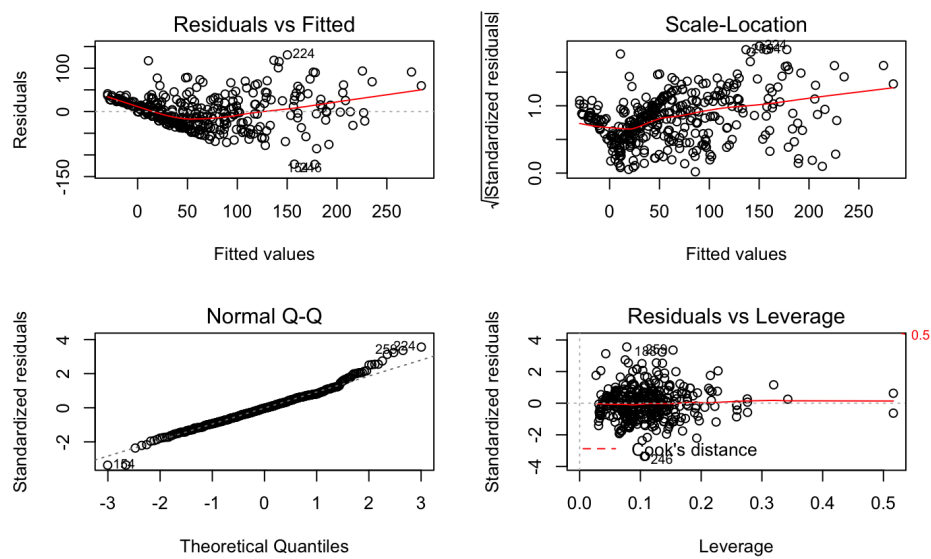


Figure 8: Regression diagnostic plot for fit. From top to bottom, left to right: Residual vs Fitted plot, Scale-Location plot, Normal Q-Q plot and Residual vs Leverage plot.

4 The impact of meteorological factors on the duration of flights

The meteorological data of the time of take off has been collected to study the impact of meteorological factors on the duration of flight of the gliders. First, a filtration was needed since a lot of flights were for purpose of training or check up on the equipments, so only flights of over 45 minutes have been considered.

In addition, the wind direction was categorised in a simplified way according to typical Swiss weather situations (1-90°: Bise, 91-180°: Föhn, 181-270°: westerly wind, 271-360°: Joran). Several linear regression model were built to find which factors were crucial in order to have a long flight. Using hypothesis testing, we used backward selection to eliminate the ones of high collinearity, and the ones that are not significant and also to maintain the log-likelihood side by side.

Table 1 is the different values of the coefficients that can be observed from a linear regression with all the meteorological variables. Lesser the p-value (fifth column) is for each beta of the variable, more significant the variable itself is. According to the table, apart from the wind peak gust the variables do not show much significance : it is not possible to reject the null hypothesis, which is to not include the variable in the model with a reasonable high confidence interval. In addition, the condition number of this model is very large (4.03e+20), which means there is a high collinearity between the variables. Therefore it is possible to have a model with less variables and a similar accuracy.

β	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	-1065.4974	1620.512	-0.658	0.512
wdir_cat[NE]	34.7061	23.337	1.487	0.139
wdir_cat[SW]	-2.1221	14.464	-0.147	0.884
wdir_cat[WN]	-34.7693	32.796	-1.060	0.291
temp	12.6387	12.221	1.034	0.303
dwpt	-14.0262	13.311	-1.054	0.294
rhum	4.4849	4.239	1.058	0.292
prcp	2.087e-12	3.18e-12	0.656	0.513
wspd	-2.3592	1.718	1.373	0.172
wpgt	-2.9812	0.960	-3.106	0.002
pres	0.8632	1.607	0.537	0.592
Log-Likelihood	-885.29			
Cond. num	4.03e+20			

Table 1: Summary of the parameter estimation with all the meteorological variables

By evaluating the Pearson coefficient between the meteorological variables, it was possible to eliminate some variables that are very dependent to each other. For example, the wind speed had a high correlation with the wind peak gust (0.91), the dew point temperature and the temperature are also highly correlated (0.89). In addition, the use of categories of wind direction instead of the actual value has been confirmed to be useful, since the collinearity is very high (0.96). Then, the variables with a high p-value can also be eliminated (temperature, humidity, precipitation, pressure). The wind direction remained, because by doing a regression with the actual value of the wind direction instead of the categories, the p-value of it was small enough to be significant. By the end what remained was only the category of the wind and the wind peak gust.

Table 2 is the final chosen regression model. An efficient model could be built out of backward selection. The condition number became significantly small, and with a loss of log-likelihood equal to 1.96. The latter shows that the model is still accurate with a difference of degree of freedom of 6.

β	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	162.4627	14.883	10.916	0.000
wdir_cat[NE]	47.231	20.434	2.311	0.022
wdir_cat[SW]	-1.6713	13.750	-0.122	0.903
wdir_cat[WN]	-15.0788	26.858	-0.561	0.575
wpgt	-1.9358	0.564	-3.433	0.001
Log-Likelihood	-887.25			
Cond. num	170			

Table 2: Summary of the parameter estimation of the final model

Another interesting fact to discover is the impact of wind directions on the duration of the flight. The Föhn has been normalised in this model, and compared to it, the Bise is favorable to the duration of flight whereas the two others are not.

In a nutshell, the information about the wind peak gust and the wind direction was sufficient to have an accurate model of the duration of flight of the gliders with respect to meteorological conditions. An exploration of better models using quasi-Poisson regression or negative-binomial regression can be done to ameliorate the accuracy.

5 Towing performance

5.1 Towing time optimization

In this part, the aim is to optimise the towing time for long flights, i.e. to build a model predicting an optimal duration according to the current flight parameters. To achieve this goal, a suitable method would be to perform linear regressions.

As a first step, it is necessary to exclude variables that do not influence the towing time or whose usefulness does not make sense in a regression problem. The variables that are not considered are the date, day, month, type of flight, landing times and whether the glider is part of the GVVN or is private.

Second, some variables have to be categorised so that they can be used for linear regression. The takeoff time was replaced by 0 if it was before noon, and by 1 if the flight started afterwards.

The flights of interest here are the long-duration flights, i.e. all other flights whose purpose was not to stay in the air as long as possible (such as training flights, acrobatic flights, check flights, etc.) should be removed. From here on, two sub-samples are considered (long and short flights). Based on personal experience, a long flight is considered to last at least 45 minutes. Of these flights, five were removed because the towing time was not indicated. In three cases, it was a private autonomous glider and in the other two cases it was missing values. Seven flights per sub-sample were again removed according to the IQR¹ criterion, as

¹IQR stands for interquartile range (measure of statistical dispersion).

they were considered as outliers. This is the same criterion used in the boxplots, i.e. a value is considered as an outlier if it lies outside the interval $[Q_1 - 1.5 \cdot \text{IQR}, Q_3 + 1.5 \cdot \text{IQR}]$. This criterion coincides with personal experience as there are no more towing times longer than 30 minutes for long flights, or longer than 20 minutes for short flights. This step is necessary because in the case of short flights the pilot must have tried to hold on but failed, while in the case of long flights the pilot may have wanted to be towed to a remote location such as the Alps. Table 3 represents an overview of the distribution of towing times for these two subsets:

Subset	N	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Long flights	155	4	10	12	12.6	16	24
Short flights	192	2	5	6	7.15	9	16

Table 3: Statistics about the towing time with respect to both subsets (long and short flights)

On average, a flight longer than 45 minutes will require five to six minutes of extra towing time (almost the same for the median). A confidence interval (at a 95% confidence level) for the average tow time of long flights is $[12, 13.2]$, i.e. the "true" mean (by taking into account all flights, not necessarily those in the year 2021) should lie within this interval with a probability of 95%.

Naturally, the tow times for long flights should be distributed according to a normal law, with a concentration towards the mean, and a progressive decrease on both sides (experienced pilots will tend to release the cable earlier, while beginners will prefer to wait a little longer to be sure to find good lift). Figure 9 confirms this intuition.

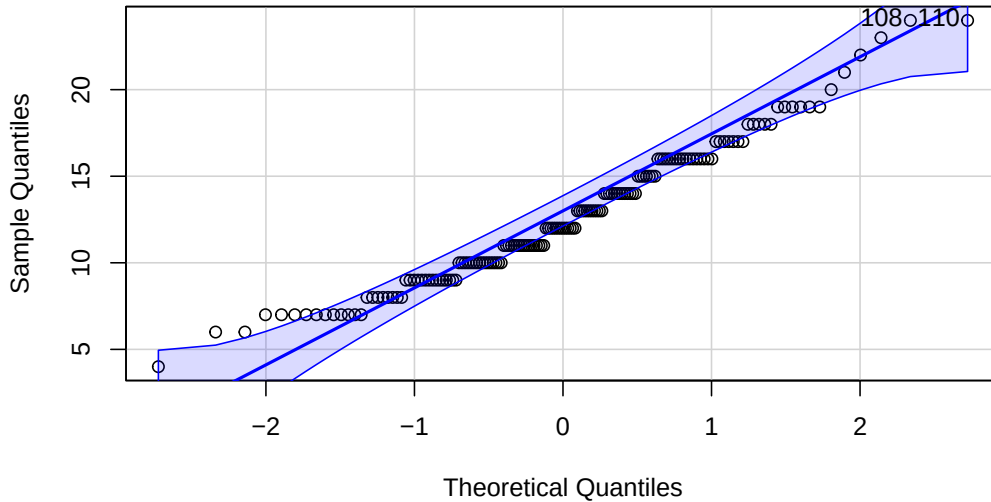


Figure 9: Q-Q plot of the towing times in the subset of flights longer than 45 minutes with a 95% confidence interval

To find a model that best predicts the towing time, a suitable approach is to perform a stepwise regression in both directions based on the AIC¹ criterion. This is an automatic procedure that

¹The Akaike information criterion gives a measure assessing the quality of a model based on the goodness of fit and the simplicity (number of independent variables involved).

adds and removes variables to the model, while evaluating the quality of the latter based on some criterion, such as the AIC. If there is a significant improvement at any stage, the variable will be removed or added depending on the direction of the improvement. The model returned is the following:

$$\text{TOW} = \beta_{\text{intercept}} + \beta_{\text{temp}} \cdot \text{temp} + \beta_{\text{wpgt}} \cdot \text{wpgt} + \beta_{\text{Planeur_xxxx}},$$

where the estimates are given in Table 4.

β	Estimate	Std. Error	t value	$\text{Pr}(> t)$	
(Intercept)	10.21	1.43	7.15	0.00	*
temp	0.21	0.05	4.49	0.00	*
wpgt	-0.09	0.02	-4.61	0.00	*
Planeur1599	-1.00	1.22	-0.82	0.41	
Planeur1636	0.33	1.07	0.31	0.76	
Planeur1840	-0.27	1.07	-0.26	0.80	
Planeur2138	1.41	2.11	0.67	0.51	
Planeur2492	1.78	1.11	1.60	0.11	
Planeur2509	0.18	1.05	0.17	0.87	
Planeur2519	-0.71	1.19	-0.60	0.55	
Planeur3185	2.69	1.05	2.57	0.01	*
Planeur3354	2.87	0.94	3.06	0.00	*
Planeur552	1.60	1.82	0.88	0.38	
Planeur653	1.92	1.59	1.21	0.23	

Table 4: Summary of the parameter estimation

This model indicates that the towing time depends on the temperature, the wind peak gust and the glider. It would be logical for the wind to be included in the model, but it turns out that the peak gust variable captures enough information about the wind movements. It should also be noted that these measurements are at the level of the city of Neuchâtel, but the wind behaviour changes with altitude. β_{temp} is positive, so an increase of temperature leads to an increase of the towing time. This reflects the situation in summer when the heat causes the air to expand, reducing lift. β_{wpgt} is negative, so an increase of the wind peak gust leads to a decrease of the towing time. This can be explained by the fact that gusts induce wind. The tow plane generally flies into the wind to optimise the climb, thus reducing the time the windier it is. These two coefficients are all significantly different from zero, which is not the case for every glider coefficient. This is probably due to the fact that there is less data for some gliders. Typically, the 552 and 653 are vintage gliders that rarely make long flights (because of their lower performance). Gliders 3185 and 3354 have significant coefficients. These are two-seaters, so it is logical to add two to three minutes of additional towing time for these types of glider. The coefficient that has been normalised is that of the 1419 glider ($\beta_{\text{Planeur_1419}} = 0$). Note that all gliders that have taken off and landed in Neuchâtel are taken into account, including private gliders.

The diagnostic plots for checking the regression assumptions are shown in Figure 10. The first plot of residuals vs. fitted values confirms the hypothesis of linearity between the predictor and the outcome. The curve is barely raised on the left, but this is mainly due to

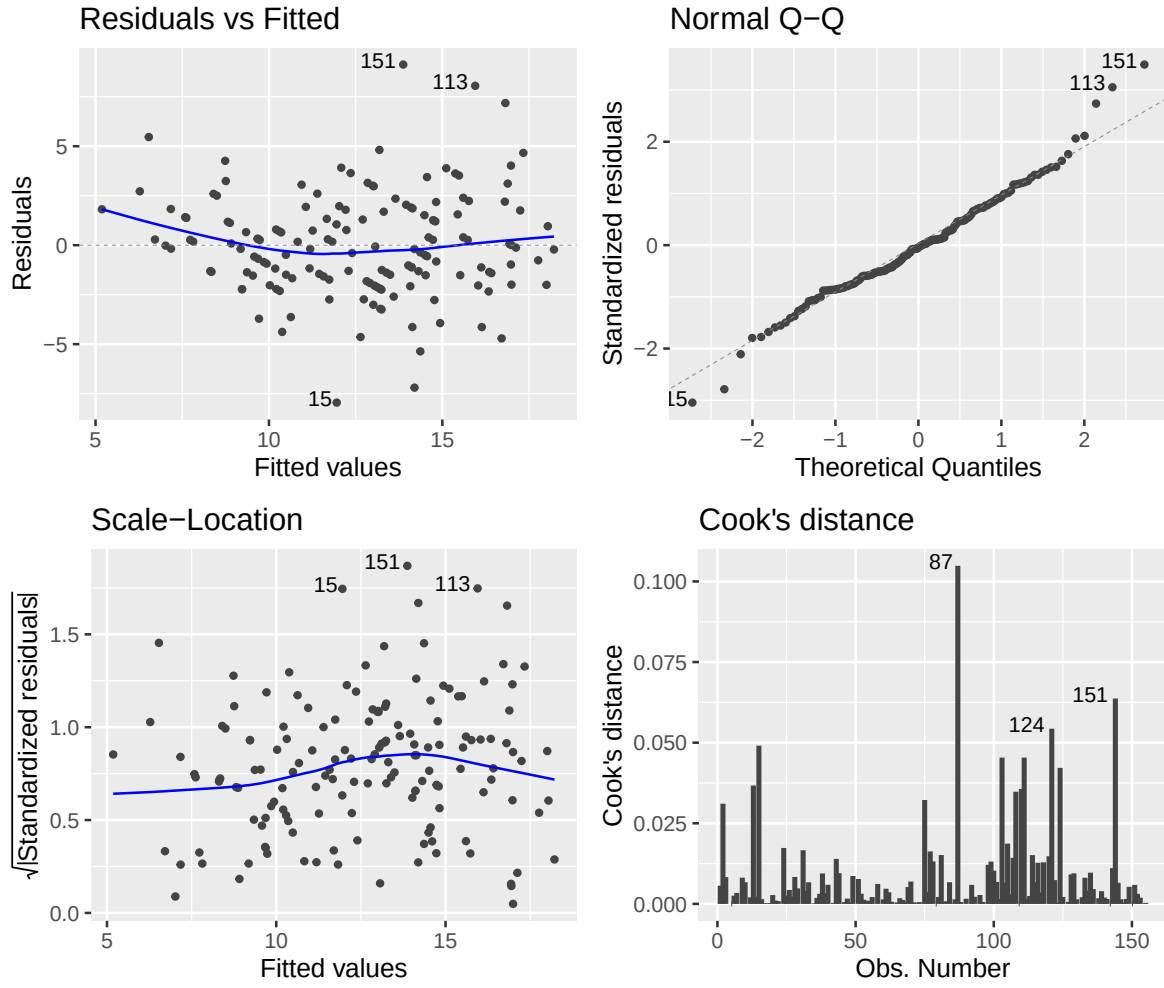


Figure 10: Diagnostic plots of the linear model

the fact that there is less data in this region, as flights longer than 45 minutes are relatively rare for a towing time of five to seven minutes. For the few observations highlighted (15, 113 and 151), these are the most extreme tow times retained (4, 24 and 23 minutes respectively). The Q-Q plot shows that the residuals are normally distributed. Again, the same more extreme values will be more difficult to predict, hence they have more difficulty in aligning with the other points.

The Scale-Location plot displays the homogeneity of the variance of the residuals (homoscedasticity). Once again, the three extreme values slightly disturb the curve.

As for the influential values, observation 151 is highlighted for the same reasons mentioned above. Observation 124 corresponds to a day of strong Bise (27 km/h wind), which will influence the coefficient β_{wpgt} . Finally, observation 87 refers to the flight of glider 653 (oldtimer) in normal conditions. As this glider rarely makes flights longer than 45 minutes, this observation probably tends to influence its coefficient $\beta_{\text{Planeur_653}}$.

Now that the validity of the model has been verified, its accuracy must be assessed. Figure 11 is a plot of fitted towing times against observed ones.

At a glance, two thirds of the observations seems to be predicted with a margin of error of plus or minus two minutes. However, much of the variation remains unexplained by the proposed

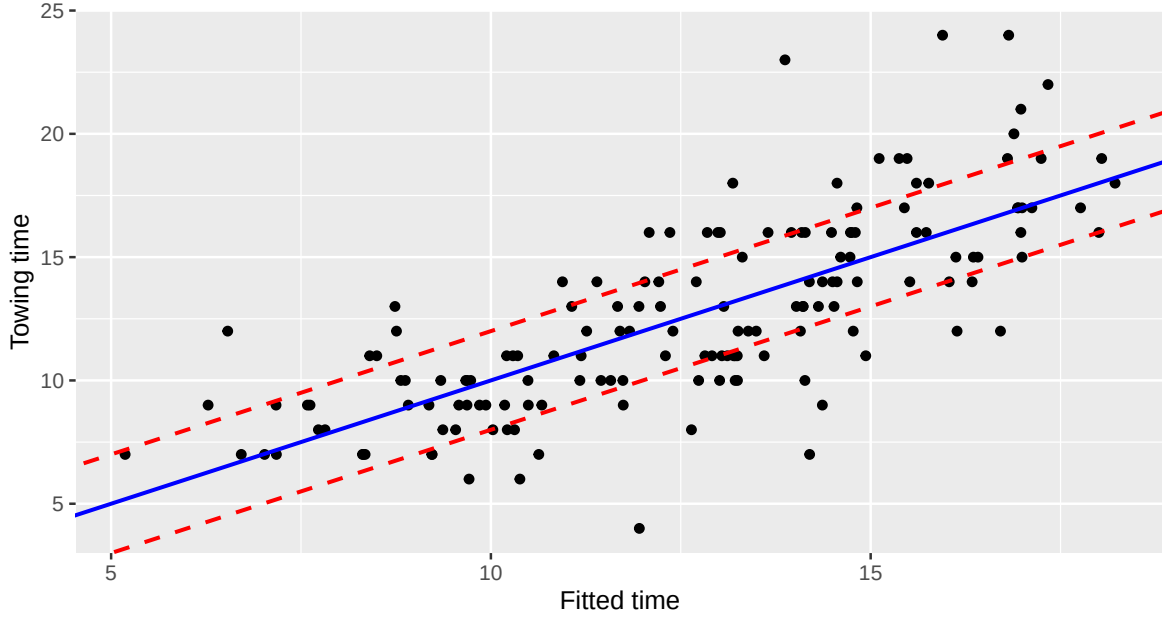


Figure 11: Fitted towing times against observed ones where the blue line has slope one with zero intercept. The red dashed lines represent margins of plus or minus two minutes.

model. The adjusted R^2 value¹ obtained is 0.53, which indicates that slightly more than half of the variation in towing time is explained. This is because this variable is difficult to predict, even if more parameters (including more specific meteorological measurements related to gliding) would have been taken into account. In this context, only observed variables are used, but this problem depends very much on the psychology of the pilots (decision making to release the cable) and would need to consider more complex modelling methods with models including latent variables² for a better fit.

Nevertheless, the model gives a satisfactory representation of reality and could be tested for the year 2022 and beyond. Thus, the pilot will be able to have an estimate of the necessary towing time (knowing that he plans to carry out a flight of more than 45 minutes) according to the two meteorological parameters and the glider selected.

5.2 Tow plane comparison

GVVN has two tow plane at disposal, WYZ and 2451. We would like to know if one is better than the other for towing duration. This analysis is made on flights that were shorter than 10 minutes in order to do it on those that have the same purpose and when the elevation objective is the same for the release of the glider.

For those flights, we first see that the ultra light aircraft (WYZ) was more often used than the touring motor glider (2451), respectively 53 times to 29. Thus, we would like to know if the ultra light aircraft is significantly faster than the other one.

Table 5 is a summary of the distribution of towing time with respect to both tow plane.

¹Statistical measure of fit between zero and one that indicates how much variation of a dependent variable is explained by the independent variable(s) in a regression model

²Latent variables are opposed to observed variables. They are not directly observed, but rather inferred

Subset	N	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2451	29	3	4	5	4.97	6	9
WYZ	53	3	3	4	4.49	6	7

Table 5: Statistics about the towing time with respect to the tow plane

Since the dependent variable is continuous and the independent one is categorical, we perform a 1-way ANOVA to compare difference in means. Table 6 is the summary of this analysis. At a significance level of $\alpha = 0.05$, we cannot reject the null hypothesis of equal means. Thus, we cannot conclude that the towing time is significantly different between tow plane.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Tow plane	1	4.23	4.23	2.35	0.13
Residuals	80	144.21	1.80		

Table 6: 1-way ANOVA summary: Tow plane comparison

Appendix B.1 show design plot, boxplot and design plot of the data studied, respectively of the 1-way ANOVA. All lead to confirm that our assumptions are verified and that the conclusion make sense.

5.3 Single versus two-seaters

Now, we are interested into single and two-seater gliders. Does it take longer to tow a two-seater than a single one? One could reasonably think that there could be a difference between the two due to a possible difference in weight. However, note that we can still find gliders with two seats that are lighter than some with a single seat.

As previously, this analysis is made on flights with a short flight time, less than 10 minutes. This choice is made in order to focus on towing that have the same purpose, e.g. school flight, training, etc. Moreover, we exclude from the analysis private gliders since we have no information if there are two or one-seaters, and vintage gliders since those are not the ones in which we are interested.

Table 7 is the summary of the distribution of towing times for two type of gliders.

Subset	N	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Single-seaters	12	3	3	4	3.92	4.25	6
Two-seaters	46	3	4	5	4.63	5	7

Table 7: Statistics about the towing time with respect to the number of seat

We can see that there is a slight difference in means, is this difference significant? In order to answer, we make a 1-way ANOVA with our variables. Table 8 is the summary of this analysis.

from other observed variables.

At a significance level of 0.05 we can reject the null hypothesis. Thus we can conclude that the difference in means is significant.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Seat	1	4.85	4.85	4.55	0.04
Residuals	56	59.63	1.06		

Table 8: 1-way ANOVA summary: Single and two-seaters comparison

Appendix B.2 show design plot, boxplot and design plot of the data studied, respectively of the 1-way ANOVA. These point out that our assumptions are respected and that the conclusion is justified.

6 Difference between the gliders

6.1 Do gliders fly equally ?

There are two categories of aircrafts that took off at 2021 : one that are from the club which has a registration number from GVVN, and the others that are private. Some registered flight do not have a towing duration because the gliders used can fly autonomously, and others do not have the duration of flight because the glider is from an external source, which means the time of landing was not registered at the center of Neuchâtel since it happened somewhere else. Those flights have been discarded in this part in order to compare the gliders correctly.

The first question to ask is, if the gliders flew equally for the past year. Just by counting the number of flights for each glider (Figure 12), a unbalanced trend of flights have been observed between the gliders. The 3354 took nearly the majority of the shares of flights, while 3147 only flew once throughout the whole year. This shows that there was a preference on the choice of the gliders. The gliders of the following figures (Figure 12 to 14) have been ordered as a function of the number of flights.

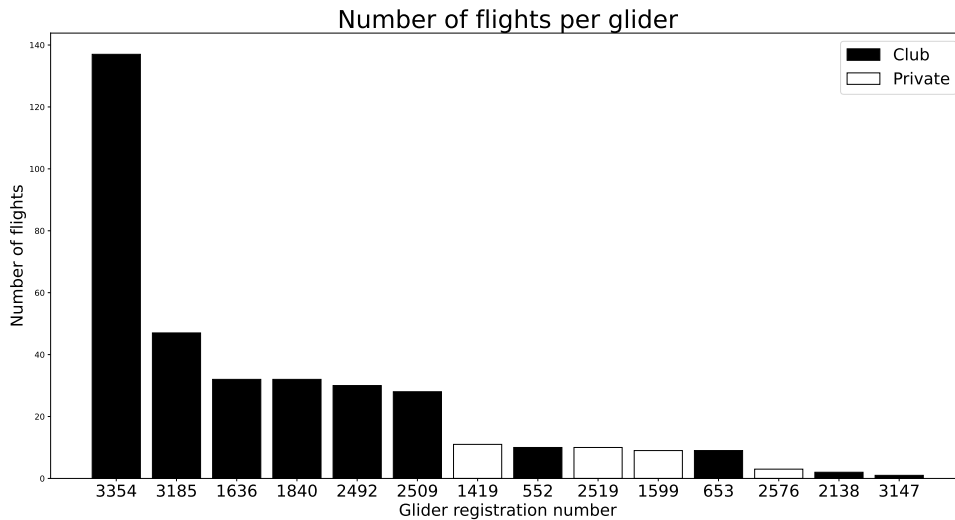


Figure 12: Number of flights per glider

Next, the comparison of the duration of towing side by side with the comparison of duration of flight is crucial to check if gliders performed equally. Figure 13 and 14 shows the average duration of towing and flights per glider. An interesting aspect is that the duration of towing does not vary as much as the number of flights. Apart from the ones that had a poor number of flights, most of them have a duration of around 10 minutes. However, the duration of flights is very dispersed between the gliders. The private glider 2519 have a overwhelming average of over 200 minutes, while the 3354 one which flew the most only has an average of about 30.55 minutes. Several reasons could explain this asymmetry : for example, the pilots might have used 3354 mostly to do a check-up or easy short flights, not to do the longest flights possible.

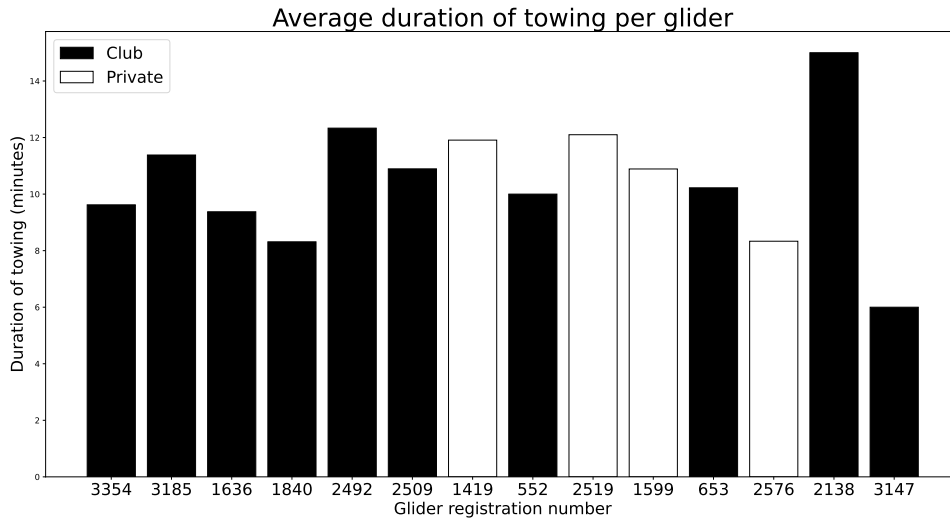


Figure 13: Average duration of towing per glider

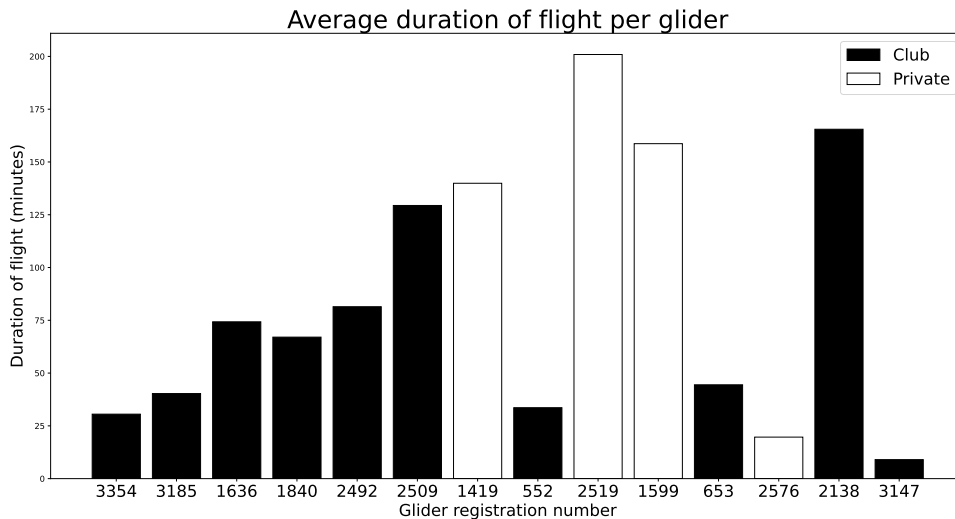


Figure 14: Average duration of flight per glider

To go more into details, the same plots have been traced for flights over 45 minutes. Figure

15 shows the proportion of number of flights that surpassed 45 minutes per glider out of all. Compared to Figure 12, some gliders have disappeared from the graph, and the rank changed. The following ones (3185, 2509, 1636, 2492, 1840) have more or less the same number of flights for both flights in general and ones of over 45 minutes. Figure 16 shows the proportion of flights longer than 45 minutes out of all flights per glider. First, 3354 has a very low percentage of long flights, which means that indeed it could have been used for other purposes. On the contrary, 2509 is probable to be one of the main ones used to aim for long flights. Second, the private gliders all flew over 45 minutes, which shows they mostly aimed to fly as long as they can as well.

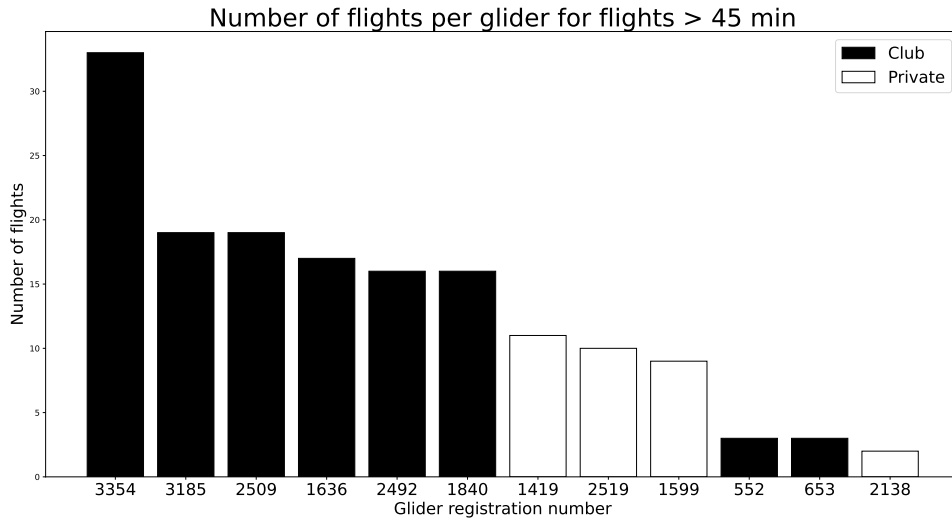


Figure 15: Number of flights per glider longer than 45 minutes

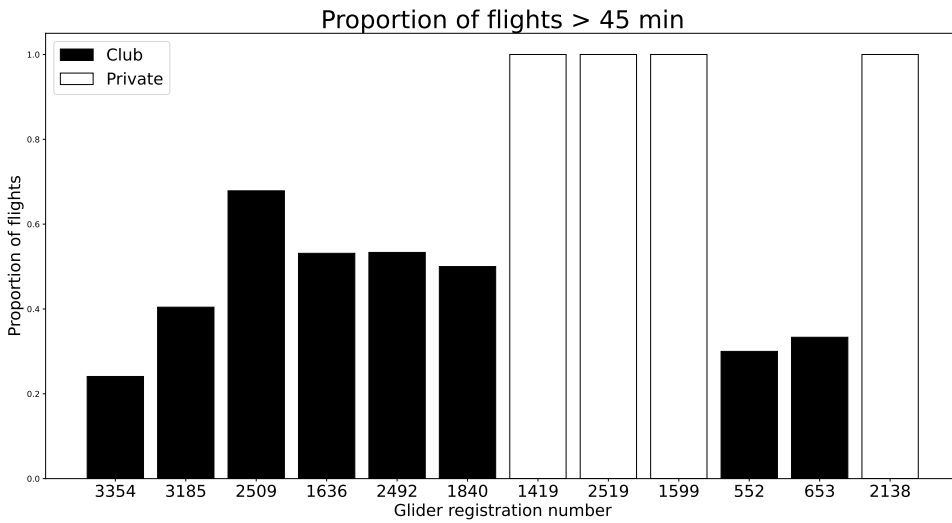


Figure 16: Proportion of number of flights longer than 45 minutes per glider

Figure 17 and 18 are the barplots of average duration of towing and flights for flights of over

45 minutes, ordered in the same way as Figure 15. As observed in Figure 13, the average of duration of towing is similar to each other on one side. On the other side, the glider 2492 had to tow longer compared to others, which could potentially mean that does not have a good performance for longer flights. On the side of the duration of flights, the variance between the gliders is smaller than the one of Figure 14, but some still stayed low overall (for instance 3354, 3185).

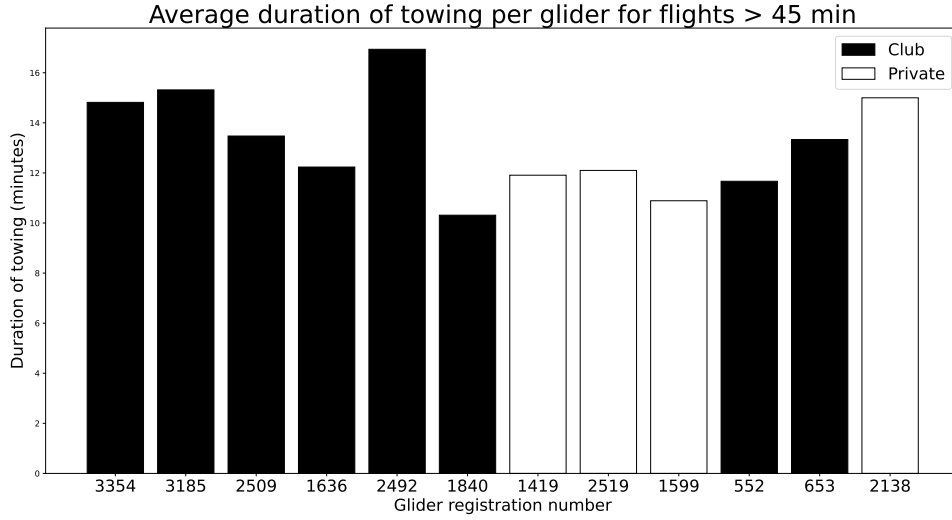


Figure 17: Average duration of towing per glider for flights longer than 45 minutes

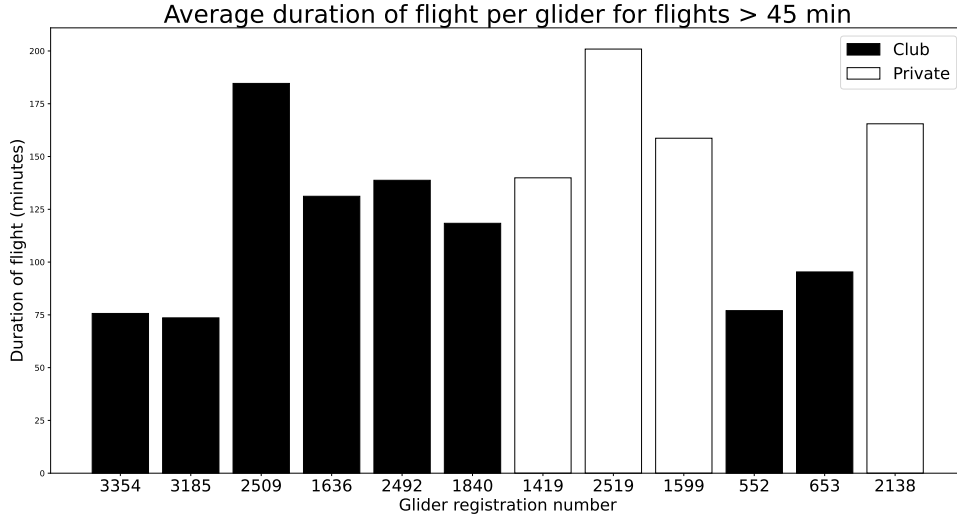


Figure 18: Average duration of flight per glider for flights longer than 45 minutes

To conclude, the gliders had a different number of flights. Although the average of duration of towing was similar to each other, the one of duration of flights had a high variance. As having a such low performance and a high performance gliders at the same time is not expected, it is possible that different gliders have different "main function", which is for example, some of

them are mostly flown for a certain purpose like doing a checkup. From the figures, the glides did not fly equally, but we cannot do a performance analysis on the gliders yet. Therefore, it is advisable to do a deeper analysis in this field, that is to determine the purpose of flights and eventually only remain the ones that were really of a purpose of long flights.

6.2 Private gliders

Some people do not use gliders from GVVN but prefer to use their own, which are private gliders. We are interested into the flight times of private gliders, but more particularly, to know if there is a significant difference in flight duration between gliders from the club and private ones.

Most of private flights purposes are not in training but more pleasure flights, thus most of private flights are longer than 45 minutes. Therefore, in this section we will work with a subset of flights that are longer than 45 minutes and for which tow time is shorter than 30 minutes (as in Section 5.1). Moreover, we exclude from the analysis motorized gliders, two from the club (2492 and 2509) and two private ones (2519 and 2576), since these gliders can obviously fly longer than unmotorized ones and are not our population of interest (in this section).

Table 9 represents a summary of the distribution of our subset.

Subset	N	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Club	90	46	61	81.5	91.47	108.75	268
Private	22	56	92.75	165.5	149.91	191.75	249

Table 9: Statistics about the flight time with respect to club and private gliders

We can clearly see that private gliders seems to flight longer than club ones. We will make a 1-way ANOVA in order to conclude if this difference is significant or no. Table 10 is the summary of the analysis. At a significance level of 0.05, we could reject the null hypothesis of equal means. Thus, we could conclude that the difference is significant.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Glider	1	60381.46	60381.46	27.25	< 0.001
Residuals	110	243770.22	2216.09		

Table 10: 1-way ANOVA summary: Club and private gliders comparison

However, Appendix C shows that the model assumptions are violated and that we should not make any conclusion based on this model.

Further improvements in our approach of the question need to be done in order to find a model that would allow us to make a conclusion.

7 Conclusion

(To be improved)

- The duration of flight follows a power law distribution.
- Proposed a linear model to see which factors are the most influential to the duration of flights
- Proposed a linear model to help pilots in their decision of releasing the cable at the right moment to fly longer than 45 minutes.
- Tow plane does not change the towing time.
- It takes longer to tow a two-seater.
- Gliders flew differently, but measurement of their performance is not possible.
- Private flight fly for longer (not yet determined).

8 Further improvements

(To be improved)

- Try different models to study what characterises the duration of flight.
- Consider more complex modelling methods with models including latent variables for a better fit (section towing time optimization and duration of flights with meteorological factors).
- Accurate model for private and club gliders comparison.
- See if the type of flights have an influence on the length of it, see if some gliders are used.
- Redo some plots considering ink plot e.g. remove the grey background, better size of plots, days and month in the right order in the barplots, consistent color and template, etc.
- Do a regression with all the variables we have, and eventually compare with the models we built separately.
- If possible later, to have a more exact data on the towing time, since the towing time includes the time for the towing aircraft to come back to the place where it took off. Also, adding information about the age of the pilots or the year passed after obtaining the license could be useful to determine the needed condition for a pilot to be competent.

9 References

Groupe de Vol à Voile de Neuchâtel, 2022. URL <https://www.gvvn.ch/index.php>. [Accessed 04-Mar-2022].

Appendix A Summary of the regression of the flight duration, excluding meteorological variables

Variable	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.1885	8.2705	0.63	0.5309	
Jour	0.6737	0.2605	2.59	0.0101	*
D_remo	5.7309	0.4384	13.07	0.0000	*
Nom_pilote_Adam	51.7455	17.4576	2.96	0.0033	*
Nom_pilote_Amber	-97.4166	40.9798	-2.38	0.0180	*
Nom_pilote_Andrew	-52.2500	21.1291	-2.47	0.0139	*
Nom_pilote_Anthony	-74.7030	41.2182	-1.81	0.0708	
Nom_pilote_Ashley	-32.1017	7.4466	-4.31	0.0000	*
Nom_pilote_Christopher	-53.7654	14.5588	-3.69	0.0003	*
Nom_pilote_David	25.3886	12.7733	1.99	0.0477	*
Nom_pilote_Elizabeth	-37.6154	14.1170	-2.66	0.0081	*
Nom_pilote_Heather	-46.1202	14.9479	-3.09	0.0022	*
Nom_pilote_Jennifer	27.4795	10.6059	2.59	0.0100	*
Nom_pilote_Jessica	-22.7175	10.2624	-2.21	0.0275	*
Nom_pilote_John	-17.5264	11.6759	-1.50	0.1343	
Nom_pilote_Justin	144.0825	12.5119	11.52	0.0000	*
Nom_pilote_Laura	-86.8111	41.3326	-2.10	0.0364	*
Nom_pilote_Melissa	14.9365	9.5538	1.56	0.1189	
Nom_pilote_Robert	-34.5123	14.8235	-2.33	0.0205	*
Nom_pilote_Ryan	-49.4910	14.0214	-3.53	0.0005	*
Moniteur_Amy	-36.6164	22.1993	-1.65	0.1000	
Moniteur_Elizabeth	-34.8783	8.5797	-4.07	0.0001	*
Moniteur_Jason	-35.6454	13.2333	-2.69	0.0074	*
Moniteur_Joseph	-39.5322	11.8654	-3.33	0.0010	*
Moniteur_Michael	-23.9575	10.4984	-2.28	0.0231	*
Pilote_remo_Crystal	-23.5042	6.6325	-3.54	0.0005	*
Pilote_remo_Joseph	-46.5867	7.4523	-6.25	0.0000	*
Pilote_remo_Ryan	-25.9985	7.9886	-3.25	0.0013	*
Pilote_remo_Steven	-34.1214	10.6533	-3.20	0.0015	*
Pilote_remo_Tiffany	-37.7871	7.1737	-5.27	0.0000	*
GV_BON	-23.9837	10.0205	-2.39	0.0172	*
GV_VCR	-61.2494	15.0563	-4.07	0.0001	*
GV_VFO	19.7565	8.7420	2.26	0.0245	*
GV_VPP	-39.5438	27.6850	-1.43	0.1541	
Planeur_1419	58.5225	12.5799	4.65	0.0000	*
Planeur_1599	86.6252	13.6707	6.34	0.0000	*
Planeur_1636	53.2852	8.6721	6.14	0.0000	*
Planeur_1840	33.6265	8.7775	3.83	0.0002	*
Planeur_2492	44.8969	9.4504	4.75	0.0000	*
Planeur_2509	94.1913	8.7700	10.74	0.0000	*
Mois_août	-15.4391	8.1789	-1.89	0.0599	
Mois_avril	34.7777	6.4130	5.42	0.0000	*
Mois_mars	-22.5088	7.0845	-3.18	0.0016	*
GV_EXT	-72.0969	13.3512	-5.40	0.0000	*

Table 11: Summary of the linear regression with flight duration as response variable and all other columns considered as observations. The regression was performed with a step selection. This summary contains the estimate, the standard error, the t-value and finally the p-value of each variable considered for the last model.

Appendix B 1-way ANOVA

B.1 Tow plane comparison (Section 5.2)

Figure 19 shows that means of our sample is not equal but that they are quite similar.

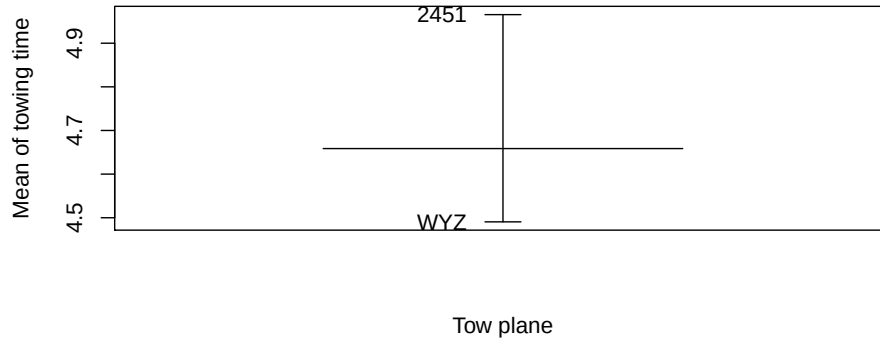


Figure 19: Design plot for tow plane comparison

Figure 20 shows that towing time of WYZ are more concentrated around the mean of our sample, conversely to towing time of 2451. This could mean that our homogeneity of variance assumption could not be true.

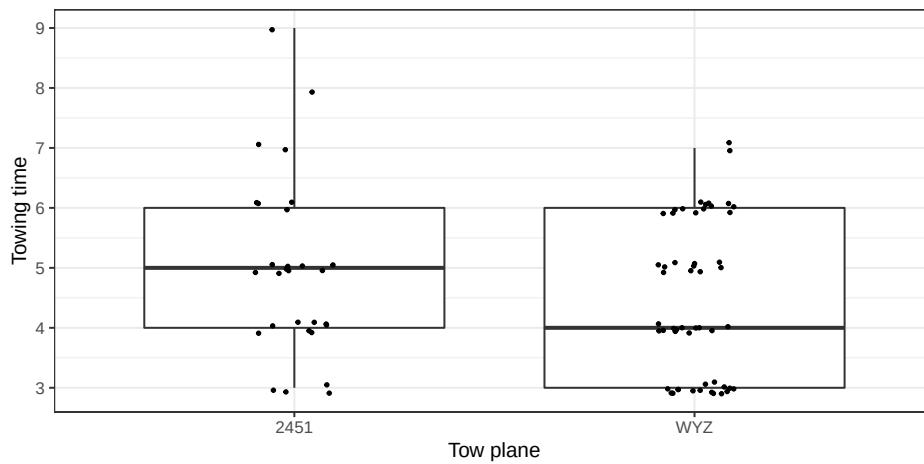


Figure 20: Boxplot for tow plane comparison

Figure 21 are diagnostic plots of our 1-way ANOVA. Thus, we can check if model assumptions are respected.

Top left plot shows most residuals are in the interval $[-2, 2]$, which satisfies homogeneity of variance assumption, except for some values (14, 55 and 64) which are some of the extreme values we have for towing time (9, 8 and 7 minutes). Bottom left plot shows that to.

Top right panel is a Q-Q plot of residuals to check the normality assumption of our data. Points seems to form a straight line which satisfies our assumption.

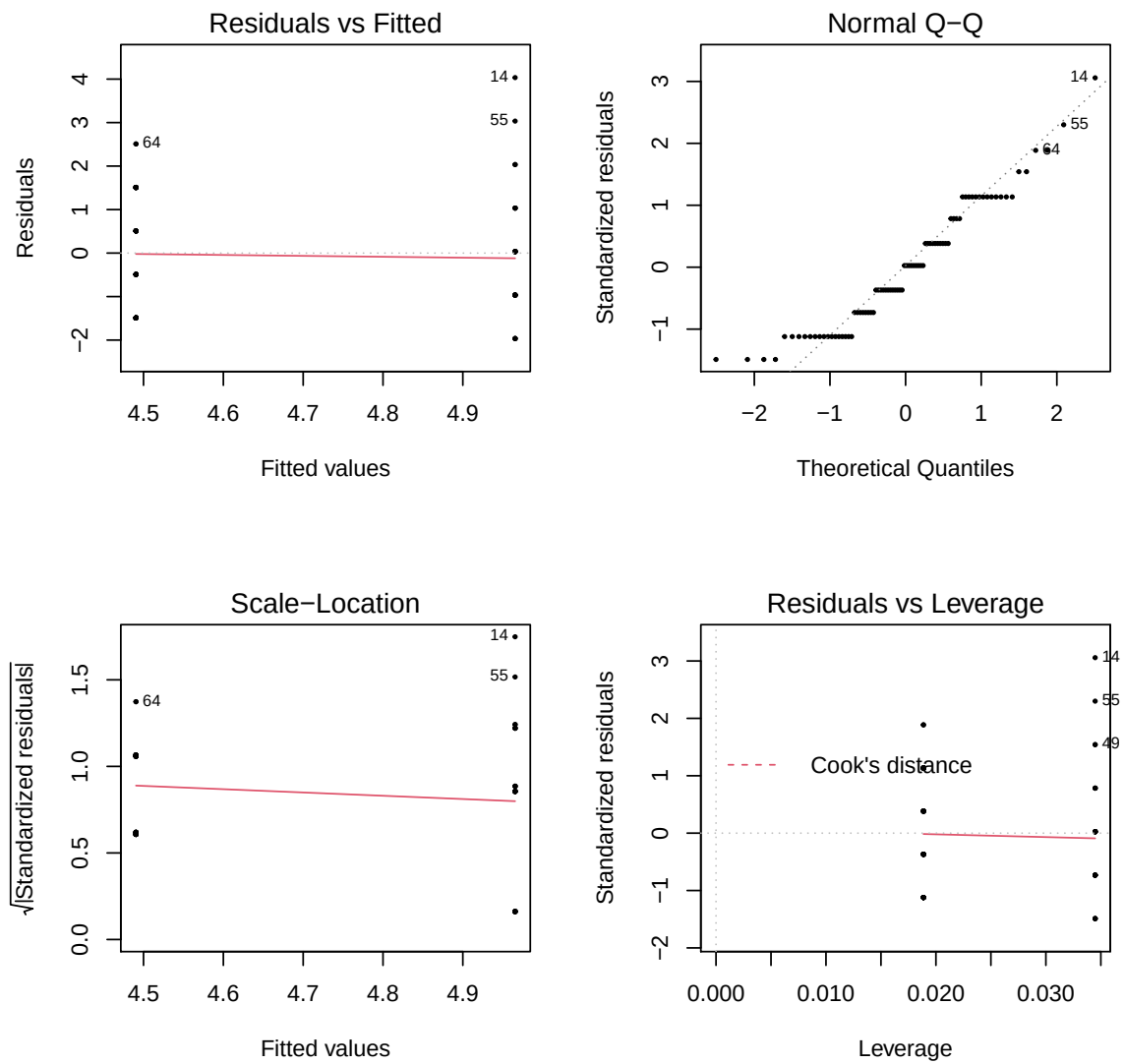


Figure 21: Diagnostic plots for tow plane comparison

B.2 Single versus two-seaters (Section 5.3)

Figure 22 shows that mean of towing time is higher for two-seaters than single-seaters.

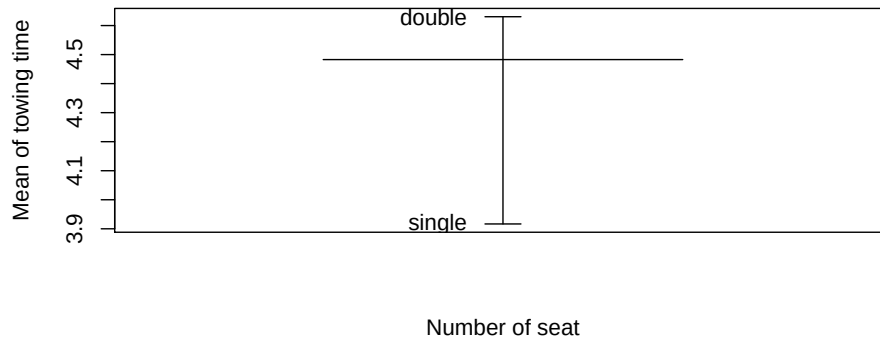


Figure 22: Design plot for single and two-seaters comparison

Figure 23 shows that the variance is similar between groups (single and double) but that normality assumption could be violated since the left box does not seem to be symmetric.

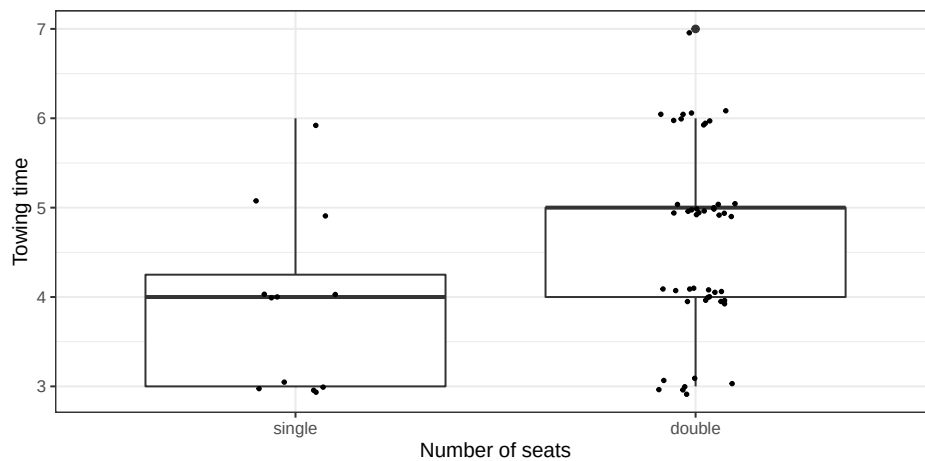


Figure 23: Boxplot for single and two-seaters comparison

Figure 24 is the diagnostic plots of our ANOVA.

Top and bottom left plots satisfies homoscedasticity assumption.

The Q-Q plot shows more heavy-tailed distribution, even if most points are near the straight line. Highlighted values (9, 50) are extremes (6 and 7 minutes), but taking a look back at the boxplot, they are not so extremes since the median is near these value.

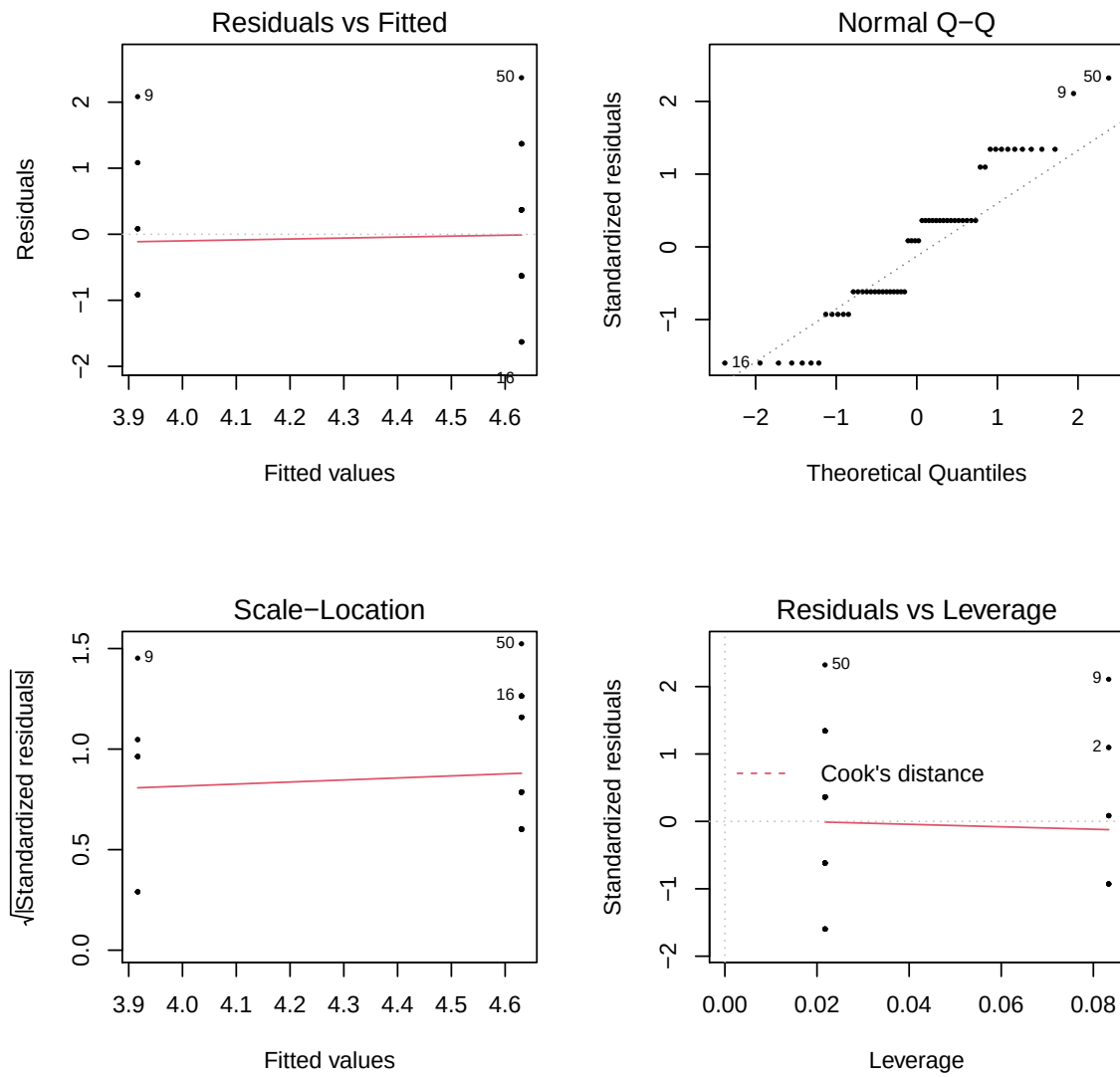


Figure 24: Diagnostic plots for single and two-seaters comparison

Appendix C Club and private gliders (Section 6.2)

Figure 25 shows us a large difference in our sample means.

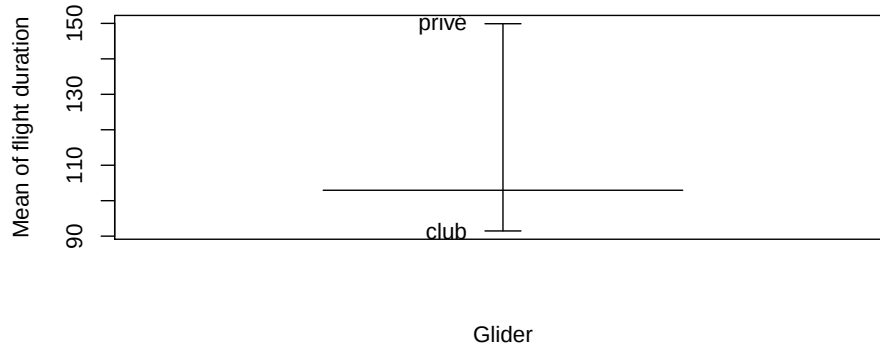


Figure 25: Design plot for club and private gliders comparison

Figure 26 shows that it seems to be a lot of outliers for the club gliders. The club box does not seem symmetric, conversely to the private one.

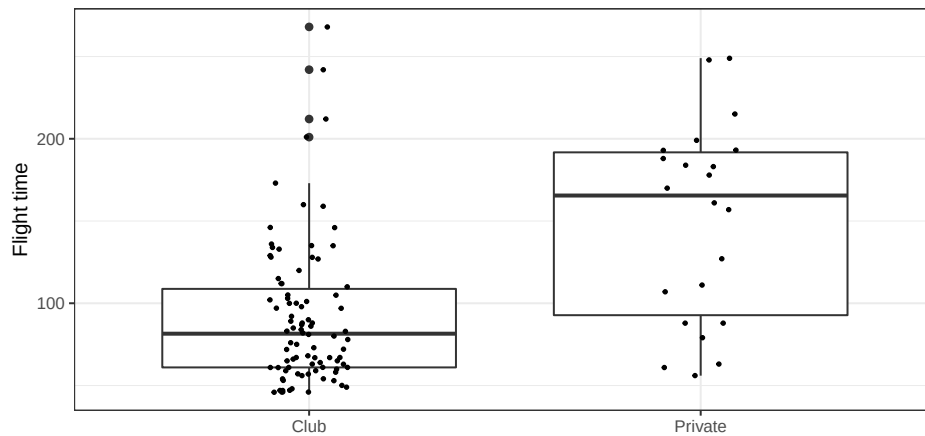


Figure 26: Boxplot for club and private gliders comparison

Figure 27 shows the diagnostic plot of our ANOVA.

Homoscedasticity assumption seems to not be respected since values are not equally dispersed around zero for each group. Other test can be performed to verify if this assumption is violated which leads to the same conclusion.

Q-Q plot seems to be skewed right by highlighted values (16, 28 and 33) which are extreme values for club gliders (242, 212 and 268 minutes) already appearing in Figure 26. Note that these values were not taken out of the final model from which we made our conclusion. These observations may signified that our data is not normally distributed.

With these observation we can conclude that the analysis is not well performed and that another type of model should be used to answer our question of interest.

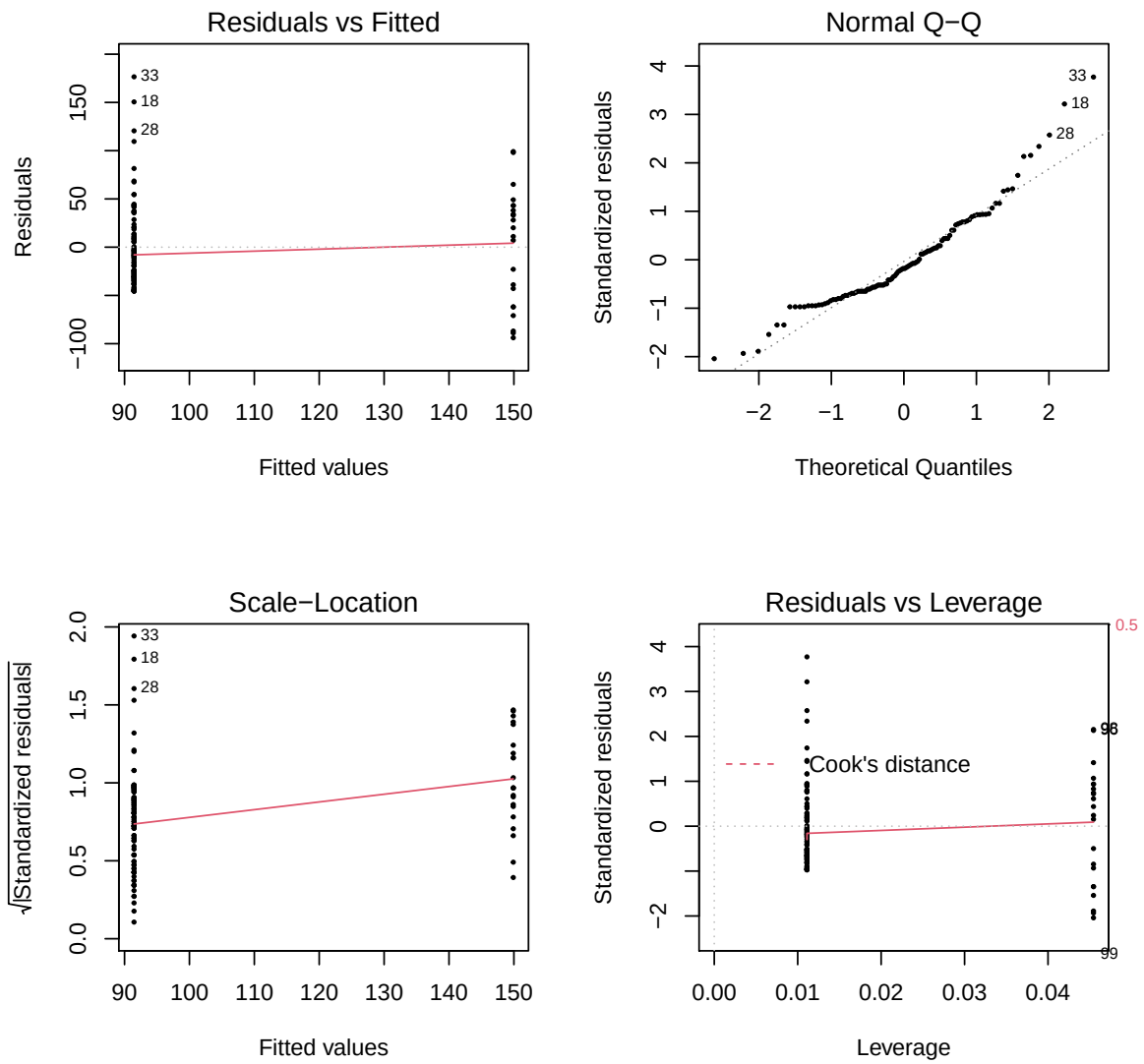


Figure 27: Diagnostic plots for club and private gliders comparison